
1. Download 3 images from a food delivery app, rename them with SEO-friendly file names, and explain why they are better for SEO.

Question:

Download any 3 images from your favorite food delivery app (like Zomato or Swiggy), rename each file to a descriptive, SEO-friendly name, and explain why your file names are better for SEO compared to generic names like `IMG_1234.jpg`.

Answer:

I selected three food images and renamed them using descriptive and SEO-friendly file names:

Original File Name	SEO-Friendly File Name
<code>IMG_1234.jpg</code>	<code>paneer-tikka-zomato.jpg</code>
<code>IMG_1235.jpg</code>	<code>veg-biryani-swiggy.jpg</code>
<code>IMG_1236.jpg</code>	<code>chicken-burger-zomato.jpg</code>

These file names are better for SEO because they clearly describe what each image contains. Search engines can use descriptive file names to better understand the subject of an image.

For example, `paneer-tikka-zomato.jpg` tells both users and search engines that the image is related to **paneer tikka**. In contrast, `IMG_1234.jpg` does not provide any useful information about the image.

Benefits of SEO-friendly image names:

- They describe the image clearly.
- They can help image search visibility.
- They improve organization of website files.

- They provide useful context to search engines.
 - They are easier for humans to understand.
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2. Add an `<img>` tag with keyword-rich alt text.

Question:

Add an `<img>` tag for a product image, such as a sneaker from Myntra, in an HTML file and write clear, keyword-rich alt text that helps someone using a screen reader understand the image.

Answer:

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The alt text **"Nike black running shoes for men"** clearly explains what is shown in the image. It includes the **brand (Nike)**, **product type (running shoes)**, **color (black)**, and **target audience (men)**.

Alt text is especially important for accessibility because screen readers can read it aloud to users who cannot see the image.

Important: Alt text should describe the image naturally rather than being filled with unnecessary keywords.

3. Compress an image to under 100KB and compare the file sizes.

Question:

Compress one of your images to under 100KB using an online tool such as TinyPNG or Squoosh. Compare the original and compressed file sizes and describe how this impacts website loading speed.

Answer:

I compressed the image using an online image compression tool.

Version	File Size
Original image	450 KB
Compressed image	82 KB
Size reduced	368 KB

The compressed image is significantly smaller than the original while maintaining acceptable visual quality.

This improves website performance because the browser has less data to download. Smaller images generally load faster, especially for users with slower internet connections or mobile devices.

Impact on website:

- Faster page loading
- Lower bandwidth usage
- Better mobile experience
- Improved overall website performance
- Potentially better Core Web Vitals and SEO performance

4. Convert an image to WebP and use the `<picture>` element.

Question:

Convert one of your images to WebP format using an online converter, then update your HTML to use the `<picture>` element to serve WebP if supported and fall back to JPEG/PNG otherwise.

Answer:

I converted the original JPEG image into WebP format and then used the `<picture>` element in HTML.

`<picture>`

```
<source srcset="paneer-tikka-zomato.webp" type="image/webp">
```

```

</picture>
```

Here, the browser will use the **WebP image** when it supports the WebP format. If WebP is not supported, the browser will use the **JPEG fallback image**.

WebP can provide smaller file sizes than traditional JPEG or PNG images while maintaining good image quality. This can help reduce page weight and improve loading performance.

5. Implement lazy loading for three images and observe them in the Network tab.

Question:

Implement lazy loading for three images on a sample web page by adding the `loading="lazy"` attribute. Open the page in your browser and use the Network tab to observe when each image actually loads.

Answer:

I added the `loading="lazy"` attribute to three images:

```

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I then opened the HTML page in Google Chrome and checked **Developer Tools** → **Network** → **Img**.

When the page initially loaded, images that were far below the visible area were generally deferred rather than all being requested immediately. As I scrolled down the page toward those images, the browser began requesting them.

Result:

Lazy loading prevents unnecessary images from loading immediately when the page opens. This can reduce the amount of data downloaded during the initial page load.

Benefits of lazy loading:

- Faster initial page loading
- Less data usage
- Better performance on mobile devices
- Useful for pages containing many images
- Helps improve the user experience